

World's First PROGRAMMABLE SMART WATCH With Fitness Tracking

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A smart watch monitors and tracks the wearer's location and health data besides showing time and date. But it usually costs ₹5,000 to 50,000 and is not programmable. The smart watch suggested here can be made for less than ₹2,000 and will perform all the basic smart watch functions. It can even be programmed for additional functions and customised according to your need to track, for instance, your blood oxygen percentage.

This smart watch is based on a Wi-Fi chip that helps to customise it as part of a mesh network. It

BILL OF MATERIAL

Components	Quantity	Description
ESP12F (MOD3)	1	SMD Wi-Fi SoC chip
MCP73812T (MOD4)	1	Battery charging IC
Round LCD (MOD1)	1	GC9A01 round LCD display
MAX 30100 (MOD2)	1	SpO2 and heart sensor
LED	1	SMD LED
Resistors: 470Ω (R2), 2kΩ (R3), 100Ω (R1)	1	SMD
Capacitor 4.7μF (C1, C2)	2	SMD ceramic
2-pin jumper (JP1)	1	Jumper pin
Micro USB	1	SMD micro USB female
Switch (S1)	1	SMD slide switch
3.3V battery	1	Microsize for watch
3D printed case	1	Body for watch



Fig. 1: Author's prototype

can thus track a large number of people's health data connected on the mesh network. There are many more functions and possibilities that can be implemented on it by programming it accordingly. The author's prototype of the smart watch is shown in Fig. 1.

Circuit diagram of the programmable smart watch is shown in Fig. 2. The watch is built around a

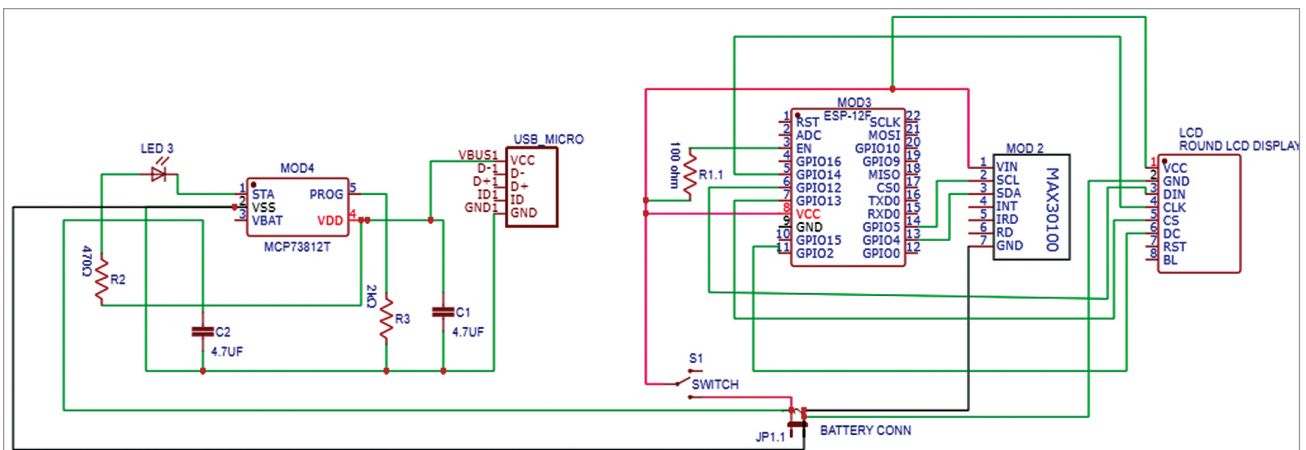


Fig. 2: Circuit diagram